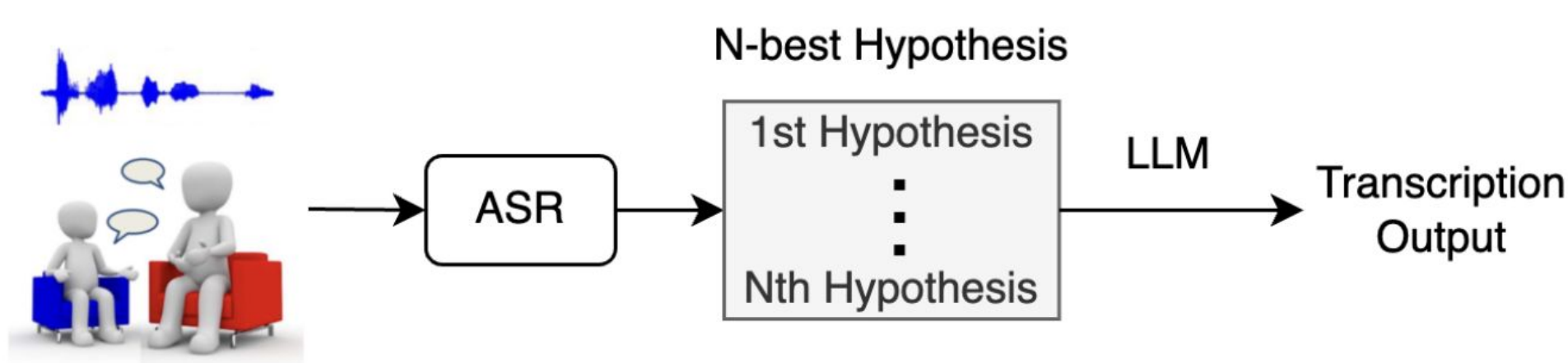


Large Language Models based ASR Error Correction for Child Conversations

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Introduction

- **ASR** is crucial in automating the **child** clinical assessments/interventions.
- ASR performances are much worse for child speech, especially in **conversations**.
- We investigate the **effectiveness of LLMs** in **improving ASR transcription**.
- Specifically, we ask LLM to correct ASR errors given 5 ASR hypothesis.



LLM for Fine-tuned ASR

- Can LLM correct finetuned ASR errors?
- ASR modes:
Whisper-large-v3-turbo (**WSP-L-T**)
WavLM-large (**WavLM-L**) with CTC finetune.
- LLM error correction **improves (CTC-based) WavLM outputs**, while it **degrades the (AED-based) Whisper outputs**.

Table 2: WER comparison with LLM for fine-tuned ASR output error correction using ADOS-Mod3 and MyST dataset.

ASR	LLaMA3	Overall	ADOS Child	Adult	MyST Child
WSP-L-T	Unused	32.11	46.99	19.47	14.31
	1B	33.25	47.93	20.77	14.55
	8B	32.92	47.47	20.56	14.40
WavLM-L	Unused	66.33	88.05	47.87	27.54
	1B	56.83	78.34	38.54	19.93
	8B	50.58	72.24	16.45	16.45

Datasets

- **MyST**: Public dataset of child and virtual tutor conversations from 8 to 12 years old. Only child-side speech is included.
- **ADOS-Mod3**: In-house dataset of 352 child and clinician interaction sessions of Autism diagnosis. Involves 180 children from 2 to 13 years olds. Both child and adult speech are included.

LLM for Zero-Shot ASR

- Can LLM correct zero-shot ASR errors?
- We show that **LLM consistently improves the zero-shot ASR** outputs across the two datasets.
- Larger LLM hws better improvements.

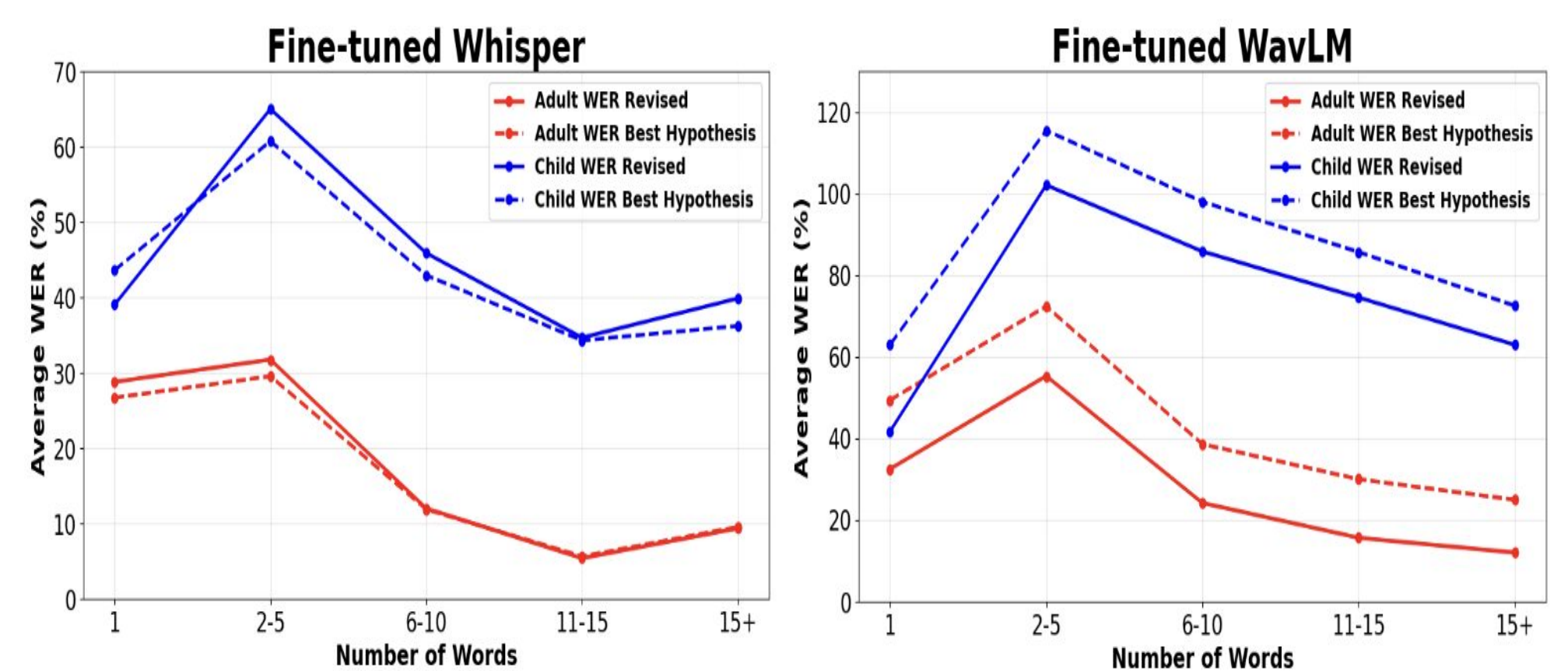
Table 1: WER comparison with LLM for zero-shot ASR error correction using ADOS-Mod3 and MyST dataset.

ASR	LLaMA3	Overall	ADOS Child	Adult	MyST Child
WSP-S	Unused	46.67	63.73	32.23	22.33
	1B	47.19	64.64	32.41	22.20
	8B	43.96	62.71	28.10	20.60
WSP-L-T	Unused	40.77	55.84	28.07	20.01
	1B	39.11	54.29	26.30	19.66
	8B	37.09	53.87	22.94	18.35
WSP-L	Unused	40.26	55.19	27.65	19.58
	1B	39.55	54.48	26.93	19.50
	8B	36.70	52.63	23.24	18.41

- Whisper-small (**WSP-S**), whisper-large-v3-turbo (**WSP-L-T**), whisper-large-v3 (**WSP-L**).
- LLM model: **LLaMA3**.

Analysis

1. Error correction performance by utterance length.



(Using ADOS-Mod-3)

2. Can conversational history improve the results?

We do this by adding ASR outputs for preceding utterances in the prompt.

We found conversational history **degrades** the LLM error correction, due to **error propagation**.

Conclusion

- We showed that LLM is effective at correcting zero-shot ASR results.
- LLM fails to improve fine-tuned AED-based ASR outputs.
- Conversational context does not Improve LLM error correction.
- Paper QR code to the right.

